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Toward quantification of flexibility for an electricity-heat integrated energy system: Secure operation region approach

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Abstract
With the tight interconnection of multiple energy flows in the integrated energy system, the security problem owing to the interaction between the electric network and the heating network has gained wide attention. To facilitate the security assessment of the integrated energy system, this article initially proposes the definition for the secure operation region (SOR) of the heat-electricity integrated energy system under the exact AC power flow model and non-linear heating formulation. The SOR of the integrated energy system is defined as a set of operation points for which the operational constraints of the electric network and the coupling heating network are satisfied. Next, complete characterization for the SOR of the integrated energy system is achieved based on the non-linear dynamical quotient gradient system. After that, the evaluation criteria based on the integrated energy system (IES) SOR is defined to screen flexibility providers. Flexibility improvement by utilizing plug-in hybrid electrical vehicle, installing energy storage, installing a heat storage tank, installing electric boiler, and utilizing building thermal inertia in the heat network is simulated and measured by application of the completely characterized IES SOR. Finally, numerical studies are given to verify the effectiveness of the proposed scheme.

1 | INTRODUCTION

The integration of multi-energy (electricity, gas, heat etc.) receives great attention from researchers and policy-makers. This integrated energy system (IES) can lead to more efficient energy utilization and secure energy supply. It can also improve the penetration of renewable energy, and promote the coordinated operation and planning of the IESs. One successful example is the combined heat and electricity system, coupling heating and electricity systems. Even though several advantages can be brought about by the IES, it may encounter some operational and planning challenges [1].

The secure operation region (SOR) is a useful tool that can enable assessment of the operational flexibility of the IES. Through the SOR, IES operators can not only assess if a given operation point is feasible, but also determines if the security margin of each credible scenario is feasible or not. However, little research effort was directed towards the SOR of the IES. Hence, the issue of a SOR of IES should be adequately addressed, in addition to the issues of the model, analysis, and optimization (see e.g. [2–4]). In light of the above considerations, this article seeks to develop a complete characterization for SORs of the electricity and heat IES benefiting the reliable and economic operation of the IES.

A great number of methods have been proposed to estimate the steady-state security region of the power system. The concept of a static security region over the transmission corridor was introduced in the mid-1970s [5]. The 'steady-state security region' was proposed in the 1970s that defines a set of real and reactive power injections (load demands and power generations) for which the power flow equations and the security constraints imposed by equipment operating limits are satisfied [6]. Since then, significant effort has been directed towards the study of secure operating-related regions. Liu proposed an expansion method to estimate the security region [7]. Zhu *et al.* proposed an expansion process to approximate the security region [8]. A DC-based SOR was constructed in [9]. Considering multiple binding transmission branch constraints, the loadability set based on the DC power flow model was presented in [10]. Extensive research was done by Yu et al. to develop an approximated security region expressed by hyperplanes and apply to large power system models [11, 12]. With some simplification in the AC power flow model, Nguyen et al. constructed a convex inner approximation of the steady-state security region [13]. A linear combination of basic polynomial functions using the Galerkin method was proposed in [14] to visualize the boundary of the static voltage stability region. The boundary of the distribution system security region was observed in [15]. A dynamical system has been utilized to completely characterize an optimal power flow feasible region [16]. Although significant research regarding the SOR for power systems has been conducted, very limited research is currently being conducted to construct the SOR for the IES. A sampling-fitting method to approximate the IES SOR was proposed in [17]. A great computation burden is required to track boundary points in all stress direction. The hyperplane approximation method was further extended to obtain the critical boundaries of the IES security region [18]. Considering the wind power uncertainties, the convex hull was used to describe the robust secure region of the integrated electrical power and natural gas systems in [19], where the security region is assumed to be convex and continuous. An overview of existing methods reveals that some approximation and/or simplification were assumed to construct the IES SOR. For instance, hyperplane approximation [18] or convex hull approximation [19] all give an over-estimated or very conservative estimation of the IES SOR.

The IES SOR can also be a powerful tool to assess the amount of renewable energy that can be integrated into the power system. The overall flexibility of the IES is evaluated by the criteria defined by the SOR of IES. However, the existing methods cannot provide an exact estimation of the complete characterization of the IES SOR. Moreover, the issue of how to quantify the flexibility improvement of the IES based on the IES SOR is not properly addressed in the existing methods. Traditionally, maximum flexibility is calculated for specific operational points [20]. A region-based method was proposed to estimate the flexibility of the IES in [21]; unfortunately, the estimation of the feasible region was not accurate and the model used was a simplified one. In [22], the feasible operation area of a CHP unit was used to evaluate the flexibility improvement of flexible heat sources, but the range of renewable energy that can be integrated (a sort of flexibility) into the power system was not available.

To fill this gap, we aim to develop a complete characterization of the IES SOR, where the operational constraints of coupled DHN are all taken into account. This complete characterization enables the exact calculation of the IES SOR. The effectiveness of several flexibility enhancement strategies on the power grid side such as the plug-in hybrid electrical vehicle and energy storage and on the heat network side such as the electrical boiler, the heating storage tank, and building thermal inertia is properly quantified by the exact calculation of the IES SOR. The contributions of this article are suggested as follows:

1) It develops an exact characterization for the IES SOR taking into account the exact AC power flow model, nonlinear heat model, and the engineering and operational constraints.
2) It proposes several evaluation criteria based on the IES SOR to evaluate the degree of operational flexibility. The flexibility improvement provides by plug-in hybrid electrical vehicle, energy storage, electrical boilers, heat storage tanks, and building thermal inertia are quantified utilizing these criteria.
3) It shows that the existing flexibility metrics only provide partial information of flexibility for specific operational points while the evaluation criteria introduced in this article fully utilize the IES SOR to assess the overall flexibility of the IES.

The remainder of this article is organized as follows: the IES problem formulation is introduced in Section 2. Then the definition of the IES secure region is presented in Section 3. A complete characterization of the IES secure operation is proposed in Section 4. Section 5 elaborates on the flexibility providers on the power grid side and heat network side. Section 6 presents the performance of the proposed method in the test system. Finally, Section 7 concludes the present work.

2 | MODEL OF THE ELECTRICITY-HEAT INTEGRATED ENERGY SYSTEM

In this section, the exact AC power flow models and heat models for the electric network and the district heating network are established. The detailed security operation constraints of IES are also presented.

We consider a power system with N buses. Let $G = \{1,..., N_G\}$ and $L = \{N_G + 1,\ldots, N\}$ represent the generator bus set and the load bus set, respectively. Here, we assume the generators and loads are not on the same bus. Bus 1 represents the slack bus, and there are N_L transmission lines.

The exact AC power flow equations and the security constraints considered are:

1) AC power flow equations

$$\begin{cases} P_{Gi} - P_{Li} - V_i \sum_{j\in i} V_j\left(G_{ij}\cos\theta_{ij} + B_{ij}\sin\theta_{ij}\right) = 0 \\ Q_{Gi} - Q_{Li} - V_i \sum_{j\in i} V_j\left(G_{ij}\sin\theta_{ij} - B_{ij}\cos\theta_{ij}\right) = 0 \end{cases} \quad i \in \{1,\ldots,N\} \tag{1}$$

2) Voltage magnitude constraints

$$V_i^{\min} \le V_i \le V_i^{\max} \quad i \in \{1,\ldots,N\} \tag{2}$$

3) Physical limitations of generators

$$\begin{cases} P_{Gi}^{\min} \le P_{Gi} \le P_{Gi}^{\max} \\ Q_{Gi}^{\min} \le Q_{Gi} \le Q_{Gi}^{\max} \end{cases} \quad i \in \{1, \ldots, N_G\} \tag{3}$$

4) Thermal limits of branch flows

$$\begin{cases} |S_t| \le S_l^{\max} \\ |S_f| \le S_l^{\max} \end{cases} \quad l \in \{1, \ldots, N_L\} \tag{4}$$

The problem variables of the AC power flow model are classified into two groups, namely the state variables $y_i \in \Re^n$, which are grouped into the column vector y_e, and the control variables $u_i \in \Re^m$, which are grouped into the column vector u_e. The state variables y_e are voltage vectors that depend on control variables. On the other hand, we view the real power generation and voltage magnitude of the PV bus, which can be adjusted (manipulated) in practice as the control variables u_e. The subscript e represents the electric power system.

Formally, the state and control variables can be related through Equations (1)–(4), which are compactly represented in the following equations:

$$\begin{cases} C_E(u_e, y_e(u_e)) = 0 \\ C_I(u_e, y_e(u_e)) \le 0 \end{cases}, \quad u_e \in \Re^{2N_G - 1}, y_e(u_e) \in \Re^{2N_B}$$

where C_E represents the AC power flow equation (1) and C_I represents the inequality constraints of (2)–(4).

By adding slack variables to the inequality constraints, we can transform inequality constraints into equality constraints as follows:

$$\begin{aligned} H_e(x_e) &= \begin{bmatrix} C_E(u_e, y_e(u_e)) \\ C_I(u_e, y_e(u_e)) + \hat{S}^2 \end{bmatrix} = 0 \\ x &= (u_e, y_e(u_e), s(u_e, y_e(u_e))) \in \Re^{6N_G + 4N_B + 2N_L - 1} \\ \hat{S}^2 &= \left(s_1^2, \ldots, s_{4N_G + 2N_B + 2N_L}^2\right)^T \end{aligned} \tag{5}$$

Hence, the set of constraint functions of the electric network can be represented as the following equalities:

$$\begin{aligned} &H_e(x_e) = 0 \quad : \Re^n \to \Re^m, \\ &n = 6N_G + 4N_B + 2N_L - 1 \\ &m = 4N_G + 4N_B + 2N_L \end{aligned}$$

We also consider a district heating network with N_H buses. Let $G_H = \{1, ..., N_{GH}\}$ and $L_H = \{N_{GH} + 1, \ldots, N_H\}$ represent the heat sources set and the heating loads set, respectively. The pipeline system is composed of N_{LH} supply and return pipelines and N_{loop} loops. The formulation of the heating system is as follows.

The continuity of mass flow is expressed in Equation (6), which describes the mass flow entering a node as being equal to the flow consumption at the node plus the mass flow leaving the node.

$$Am_l = m_{qi} \quad l \in \{1, ..., N_{LH}\}, \quad i \in \{1, ..., N_H\} \tag{6}$$

where A is the network incidence matrix that relates the nodes to the branches, m_l is the mass flow within each pipe, and m_{qi} is the mass flow through each node injected from a source or discharged to a load.

The loop pressure equation indicates the sum of the head losses in a closed loop is zero, as shown in Equation (7).

$$Bh_f = 0 \ , \quad h_f = K\dot{m}|\dot{m}| \ , \quad K = \frac{8Lf}{D^5 \rho^2 \pi^2 g} \tag{7}$$

where $B: \ N_{\text{loop}} \times N_{LH}$ is the loop incidence matrix that relates the loops to the branches, h_f is the vector of the head losses, L is the pipe length, D is the pipe diameter, ρ denotes water density, and g is gravitational acceleration. f *is* the friction factor that causes head losses.

The electricity-heat IESs are coupled by the CHP units, which can produce both heat and electricity. As heat sources, such as the extraction steam turbine, the relations between the generated electricity and heat are represented by the working area. This can be expressed mathematically in Equation (8).

$$\begin{aligned} \alpha_1 P_H + \beta_1 \varphi_H \le \gamma_1 \\ \alpha_2 P_H + \beta_2 \varphi_H \le \gamma_2 \\ \alpha_3 P_H + \beta_3 \varphi_H \le \gamma_3 \end{aligned} \tag{8}$$

where $\alpha_1, \ldots \alpha_3$, $\beta_1, \ldots \beta_3$, *and* $\gamma_1, \ldots \gamma_3$ are fitting coefficients of CHP working area S.

As shown in Equation (9), the heat power can be calculated by the product of mass flow and the difference between the supply temperature and the return temperature:

$$\varphi_i = C_p m_{qi}(T_{si} - T_{ri}) \quad i \in \{ \ 1, ..., N_H\} \tag{9}$$

where φ is the vector of the heat power consumed or supplied at each node; C_p is the specific heat of water; T_s is the supply temperature; and T_r is the outlet temperature.

The temperature of the mass flow through the pipe is calculated in Equation (10), which indicates that the drop ratio of the temperature change is exponential with the length of the pipe:

$$T_{end} = (T_{start} - T_a)e^{\frac{-\lambda L}{Cpm}} + T_a \tag{10}$$

where T_{start} and T_{end} are the temperatures at the start node and the end node of a pipe, T_a is the ambient temperature, and λ is the overall heat transfer coefficient of each pipe per unit length.

The temperature of water leaving a node with more than one incoming pipe is calculated using Equation (11) as the mixture temperature of the incoming flows:

$$\left(\sum \dot{m}_{\text{out}}\right) T_{\text{out}} = \sum \left(\dot{m}_{\text{in}} T_{\text{in}}\right) \tag{11}$$

where T_{out} is the mixture temperature of a node, $\dot{m}_{\text{out}}$ is the mass flow rate within a pipe leaving the node, T_{in} is the temperature of flow at the end of an incoming pipe, and $\dot{m}_{\text{in}}$ is the mass flow rate within a pipe coming into the node.

Additionally, the ranges of the mass flow m_l, the supply temperature T_s, the return temperature T_r, and the CHP generation Φ_i are given in Equations (12)–(15).

$$m^{\min} \le m_l \le m^{\max}, \quad l \in \{1, \ldots N_{LH}\} \tag{12}$$

$$T_s^{\min} \le T_{si} \le T_s^{\max}, \quad i \in \{1, \ldots N_H\} \tag{13}$$

$$T_r^{\min} \le T_{ri} \le T_r^{\max}, \quad i \in \{1, \ldots N_H\} \tag{14}$$

$$\varphi^{\min} \le \varphi \le \varphi^{\max}, \quad i \in \{1, \ldots, N_{GH}\} \tag{15}$$

For clarity, Equations (6)–(15) can be rewritten as follows. The control variables are denoted as u^h, including the supply temperature at each source node, the return temperature at each load node (before mixing), and mass flow at the source node. All the other variables are state variables defined by y^h, which are functions of control variables. The subscript h presents the district heating network.

$$\begin{cases} C_E\left(u_h, y_h(u_h)\right) = 0 \\ C_I\left(u_h, y_h(u_h)\right) \le 0 \end{cases}, \quad u_h \in \Re^{N_H+N_{GH}}, y_h(u_h) \in \Re^{9N_H+N_{LH}}$$

By adding slack variables into the inequality constraints, we can transform inequality constraints into equality constraints as follows:

$$H_h(x_h) = \begin{bmatrix} C_E\left(u_h, y_h(u_h)\right) \\ C_I\left(u_h, y_h(u_h)\right) + \hat{S}^2 \end{bmatrix} = 0$$
$$x = \left(u_h, y_h(u_h), s\left(u_h, y_h(u_h)\right)\right) \in \Re^{9N_H+6N_{GH}+3N_{LH}}$$
$$\hat{S}^2 = \left(s_1^2, \ldots, s_{4N_H+2N_{LH}+5N_{GH}}^2\right)$$

Hence, the formulation of the district heating network can be represented compactly as

$$H_h(x_h) = 0 \quad : \Re^n \to \Re^m,$$
$$n = 9N_H + 6N_{GH} + 3N_{LH}$$
$$m = 8N_H + 5N_{GH} + 2N_{LH} + N_{loop}$$

3 | IES SECURE OPERATION REGION

By coordinating the models of the electric network and district heating network, the formulation of the IES can be compactly represented in the following equations. The control variables and state variables for IES are summarized in Table 1.

$$H(x) = \begin{bmatrix} H_e(x_e) \\ H_h(x_h) \end{bmatrix} = \begin{bmatrix} C_E\left(u_e, y_e(u_e)\right) \\ C_I\left(u_e, y_e(u_e)\right) + \hat{S}_n^2 \\ C_E\left(u_h, y_h(u_h)\right) \\ C_I\left(u_h, y_h(u_h)\right) + \hat{S}_n^2 \end{bmatrix} = \begin{bmatrix} C_E(u, y(u)) \\ C_I(u, y(u)) + \hat{S}_n^2 \end{bmatrix} = 0$$
$$u = [u_e, u_h]^T, u_e \in \Re^{2N_G-1}, u_h \in \Re^{N_H+N_{GH}}$$
$$x = (u, y(u), s(u, y(u)))$$
$$x \in \Re^{6N_G+4N_B+2N_L-1+9N_H+6N_{GH}+3N_{LH}}$$
$$y(u_e, u_h) \in \Re^{2N_B+9N_H+N_{LH}}$$
$$\hat{S}_n^2 = \left(s_1^2, \ldots, s_{2N_B+4N_G+2N_L+4N_H+2N_{LH}+5N_{GH}}^2\right)^T \tag{16}$$

Recall the definition of security region of the electric network [6]; this work extends the definition of a SOR from the electric network to the electricity-heat IES. In this regard, the definition of the IES secure operation region can be represented as:

Definition 1 The IES secure operation region.

The IES SOR is a set of the operation points that satisfies the energy flow equations and security operation constraints (16) of IES, that is,

$$SOR = \{u \in \Re^{2N_G+N_{GH}+N_H-1} : H(x) = H(u, y(u), s(u, y(u))) = 0\}$$

TABLE 1 Control and state variables of electricity and heat networks

	Control variables	State variables
Electric networks	The real power generation	The voltage angle of the PV bus and PQ bus
	The voltage magnitude of the PV bus	The voltage magnitude of the PQ bus
Heat networks	The supply temperature at each source node	The supply temperature at each load node
	The return temperature at each load node (before mixing)	The return temperature at each source node
	The mass flow at the source node	The mass flow rate in each pipe

Next, a 4-bus electricity system coupled with the 3-bus heating system (see Figure 1) is employed to illustrate this concept. We assume that generator 1 is a CHP unit. The constraint set of this test system is summarized as follows:

$$
H_e(x_e)=\begin{bmatrix} P_{Gi}-P_{Li}-V_i\sum_{j\in i}V_j\left(G_{ij}\cos\theta_{ij}+B_{ij}\sin\theta_{ij}\right)=0 \\ Q_{Gi}-Q_{Li}-V_i\sum_{j\in i}V_j\left(G_{ij}\sin\theta_{ij}-B_{ij}\cos\theta_{ij}\right)=0 \\ V_i-V_i^{\max}+S_1{}^2 \\ V_i^{\min}-V_i+S_2{}^2 \\ P_{Gi}-P_{Gi}^{\max}+S_3{}^2 \\ P_{Gi}^{\min}-P_{Gi}+S_4{}^2 \\ Q_{Gi}-Q_{Gi}^{\max}+S_5{}^2 \\ Q_{Gi}^{\min}-Q_{Gi}+S_6{}^2 \\ S_f^2-\left(S_l^{\max}\right)^2+S_7{}^2 \\ S_t^2-\left(S_l^{\max}\right)^2+S_8{}^2 \end{bmatrix}_{26\times 1}
$$

$$
=\begin{bmatrix} C_E(u,y(u)) \\ C_I(u,y(u))+\hat{S}^2 \end{bmatrix}_{26\times 1}=0
$$

$$
u_e\in\Re^3, y_e(u_e)\in\Re^8 \ , \hat{S}^2=\left(s_1^2,\ldots,s_{24}^2\right)^T
$$

$$
x=\left(u_e,y_e(u_e),s_e(u_e,y_e(u_e))\right)\in\Re^{35}
$$

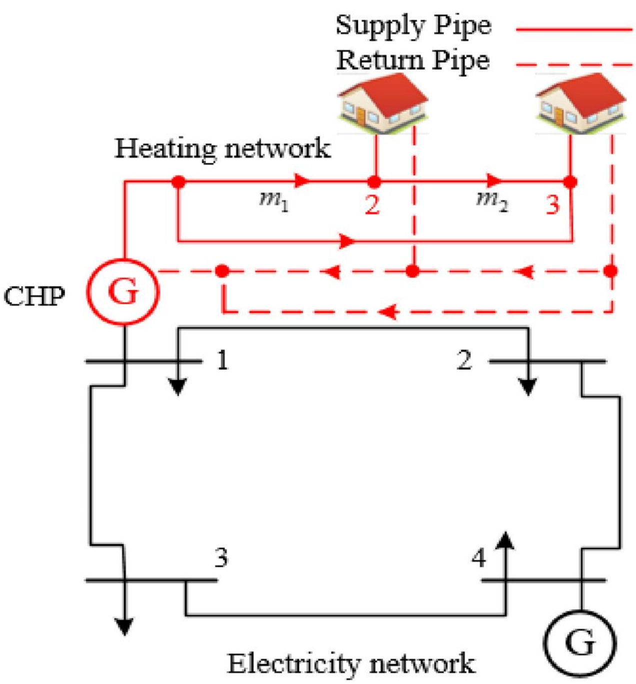

FIGURE 1 The IES 4-3 test system

$$
H_h(x_h)=\begin{bmatrix} m_{q1}-(m_1-m_2); \\ m_{q2}-(m_2+m_3); \\ m_{qs}-(-m_1-m_3)\ ; \\ K_1m_1|m_1|+K_2m_2|m_2|-K_3m_3|m_3|; \\ \phi_1=C_pm_{q1}(T_{s1}-T_{o1}); \\ \phi_2=C_pm_{q2}(T_{s2}-T_{o2}); \\ \phi_s=C_pm_{qs}(T_{ss}-T_{os}); \\ T_{s1}-(T_{ss}-T_a)e^{-\frac{\lambda L}{C_pm_1}}-T_a; \\ (m_2+m_3)T_{s2}-m_2\left((T_{s1}-T_a)e^{-\frac{\lambda L}{C_pm_2}}+T_a\right) \\ -m_3\left((T_{ss}-T_a)e^{-\frac{\lambda L}{C_pm_3}}+T_a\right); \\ m_1T_{r1}-m_2\left((T_{r2}-T_a)e^{-\frac{\lambda L}{C_pm_2}}+T_a\right)-m_{q1}T_{o1}; \\ T_{r2}-T_{o2}; \\ m_{q3}T_{rs}-m_1\left((T_{r1}-T_a)e^{-\frac{\lambda L}{C_pm_1}}+T_a\right) \\ -m_3\left((T_{r2}-T_a)e^{-\frac{\lambda L}{C_pm_3}}+T_a\right); \\ \alpha_1P_H+\beta_1\phi_H+S_1{}^2 \\ \alpha_2P_H+\beta_2\phi_H+S_2{}^2 \\ \alpha_3P_H+\beta_3\phi_H+S_3{}^2 \\ m_i-m^{\max}+S_4{}^2; \\ m^{\min}-m_i+S_5^2; \\ T_{si}-T_s{}^{\max}+S_6^2; \\ T_s{}^{\min}-T_{si}+S_7^2; \\ T_{ri}-T_r{}^{\max}+S_8^2; \\ T_r{}^{\min}-T_{ri}+S_9^2; \\ \phi_s-\phi^{\max}+S_{10}^2; \\ \phi^{\min}-\phi_s+S_{11}^2; \end{bmatrix}_{35\times 1}
$$

$$
=\begin{bmatrix} C_E(u_h,y_h(u_h)) \\ C_I(u_h,y_h(u_h))+\widehat{S}^2 \end{bmatrix}_{35\times 1}=0
$$

$$
u_h\in\Re^4, y(u_h)\in\Re^{12}, \widehat{S}^2=\left(s_1^2,\ldots,s_{23}^2\right)^T
$$

$$
x_h=\left(u_h,y_h(u_h),s(u_h,y_h(u_h))\right)\in\Re^{39},
$$

The IES SOR is shown in Figure 2. Each operating point in the IES SOR (Figure 2) represents the feasible operation point while satisfying constraints of the electric network and the coupling district heating network. For instance, the vector S_1 ($[PG_1, PG_2] = [120, 181.6]$) is a feasible power injection that can satisfy the coordination of the models of the electric network and the district heating network. Similarly, vector S_2 ($[T_{r1}, T_{r2}] = [38,35]$) is a secure return temperature of the load node that can not only guarantee the security of the heating system, but also satisfy the operating constraints of the electrical system.

4 | CHARACTERIZATION OF THE SECURE OPERATION REGION

To derive a complete characterization for the IES SOR, we design the following non-linear dynamical system associated with the equality and inequality constraints described in (16), whose steady states are explored to characterize the SOR:

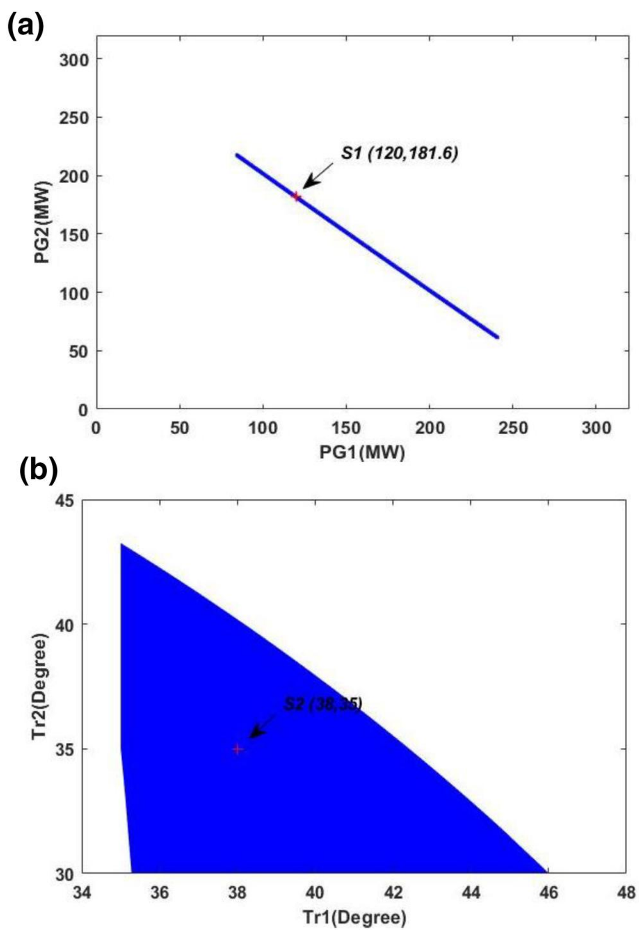

FIGURE 2 The IES secure operation region projected onto (a) the $PG_1 \times PG_2$ space and (b) the $T_{r1} \times T_{r2}$ space. Each operating point is a feasible solution while satisfying all constraints of the electric network and the district heating network

$$\dot{x} = -DH(x)^T H(x) \tag{17}$$

where $DH(x)$ is the Jacobian matrix of the vector $H(x)$ from Equation (16). The dynamical system (17) is termed the quotient gradient system (QGS), which is a non-linear, non-hyperbolic dynamical system. Due to the difference between the number of constraints and the number of variables, the steady states of the QGS are composed of equilibrium manifolds instead of equilibrium points.

Definition 2 The regular and degenerate equilibrium manifolds.

A stable equilibrium manifold (SEM) Σ_s of QGS (17) is termed a regular SEM (RSEM) if it satisfies $H(\Sigma_s) = 0$; otherwise, it is termed a degenerate SEM (DSEM).

We next derive a necessary and sufficient condition for the IES SOR.

Theorem 1: *Complete characterization.*

A path-connected set, say SOR_i, is a secure operation component of the IES model (16) if and only if it is a regular stable equilibrium manifold of the quasi-gradient system (17), that is.,

$$SOR_i = \Sigma_i^{rs} \quad , i = 1, \ldots, n$$

In addition, the SOR of the IES equals the union of the regular stable equilibrium manifold of the associated quasi-gradient system (17), that is,

$$\cup SOR_i = \cup \ \Sigma_i^{rs} \quad , i = 1, \ldots, n$$

where Σ_i^{rs} is the ith regular stable equilibrium manifold of the quasi-gradient system (17).

Proof: See the Appendix.

An important equivalence between the regular stable equilibrium manifold and the IES SOR is established in Theorem 1, which gives a complete characterization for the IES SOR. Theorem 1 is the first analytical result to give a complete characterization for the SOR of the IES. To illustrate Theorem 1, we employ the 4-bus electricity system coupled with the 3-bus heating system (see Figure 1) as an example. The corresponding QGS of the constraint set is constructed as follows:

The QGS system can be expressed as follows:

$$\dot{x} = -DH_{\mathrm{IES4\text{-}3}}(x)^T H_{\mathrm{IES4\text{-}3}}(x)$$

$$H_{\mathrm{IES4\text{-}3}}(x) \in \Re^{61} \ \ DH_{\mathrm{IES4\text{-}3}}(x) : \Re^{60\times 74}$$

$$DH_{\mathrm{IES4\text{-}3}} = \begin{bmatrix} \frac{\partial H_1}{\partial V_{a1}} & \frac{\partial H_1}{\partial V_{a2}} & \frac{\partial H_1}{\partial V_{a3}} & \cdots & \frac{\partial H_1}{\partial s_{43}} & \frac{\partial H_1}{\partial s_{44}} \\ \frac{\partial H_2}{\partial V_{a1}} & \frac{\partial H_2}{\partial V_{a2}} & \frac{\partial H_2}{\partial V_{a3}} & \cdots & \frac{\partial H_2}{\partial s_{43}} & \frac{\partial H_2}{\partial s_{44}} \\ \vdots & \ddots & \vdots & & \ddots & \vdots & \vdots \\ \frac{\partial H_{60}}{\partial V_{a1}} & \frac{\partial H_{60}}{\partial V_{a2}} & \frac{\partial H_{60}}{\partial V_{a3}} & \cdots & \frac{\partial H_{60}}{\partial s_{43}} & \frac{\partial H_{60}}{\partial s_{44}} \end{bmatrix}_{61\times 74}$$

The IES SOR (4a), (4c) and the regular stable equilibrium manifolds of the corresponding QGS (4b), (4d) are numerically computed and shown in Figure 3. The IES SOR is shown in Figure 4a. 4(c) is computed using a brute-force approach. Indeed, the IES SOR equals the regular stable equilibrium manifolds of the corresponding QGS, as asserted by Theorem 1.

Several distinguishing features of the proposed approach that differ from traditional methods are summarized in the following:

(1) The exact AC power flow models and non-linear heat models for the electric network and the district heating network are employed and can provide accurate IES operation conditions and are highly desirable for real-time operation.
(2) A complete characterization of the IES SOR based on the non-linear and non-convex model is developed by the QGS-based approach, which differs from the existing approximated SOR characterization method that constructs a series of linear programming problems, such as [21]. We note that the IES SOR characterized by the proposed method provides accurate operating information for the operators.
(3) The matrix inversion is not required for the proposed method and hence, possible ill-conditioning of the Jacobian matrix along trajectories is avoided.

5 | STRATEGIES FOR THE OPERATIONAL FLEXIBILITY ENHANCEMENT OF THE CHP UNITS

Operational flexibility is defined as the ability of power system operators to deploy adjustable resources to integrate renewable energy [20]. The operational flexibility of CHP units to reduce the power output is limited by the demand for heat supply and thus, wind power has to be curtailed. In practice, strategies to enhance the operational flexibility can be divided into two classes: the power grid side and the heat network side. Utilizing the plug-in hybrid electrical vehicle and installing energy storage can improve the flexibility in the power grid side. Three strategies in the heat network side that can alleviate the conflict between wind power integration and heat demand include installing the heat storage tank, installing the electric boiler, and utilizing building thermal inertia. These strategies have different impacts in improving the flexibility of the IES.

In this section, we apply the SOR constructed by our proposed method to examine the five strategies regarding their effectiveness in enhancing the flexibility using the following performance metric. For a given operating condition, the flexibility that can be enhanced is reflected on the expansion of the IES SOR. The larger the IES SOR, the higher the operational flexibility of the IES. To quantify the improvement of

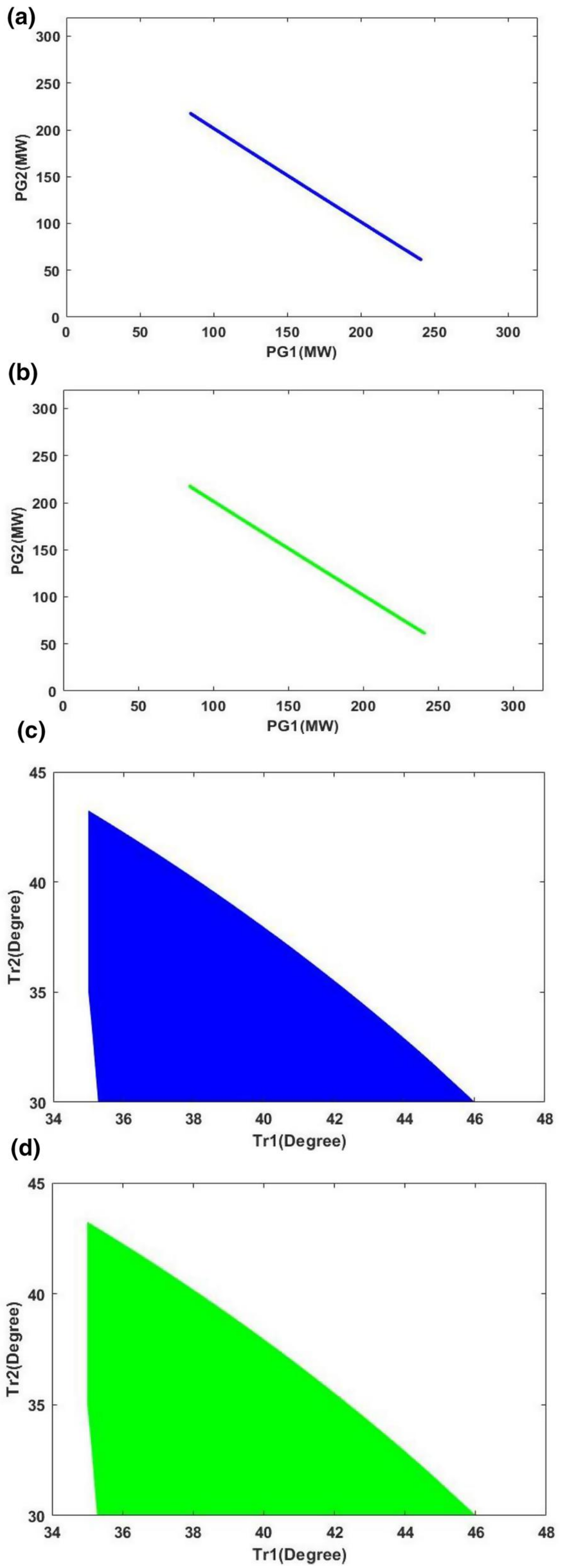

FIGURE 3 The relationship between the IES secure operation region (a, c) and the regular stable equilibrium manifolds of the IES 4-3 test system (b, d). Indeed, the regular stable equilibrium manifolds of the corresponding QGS equal the IES secure operation region, as asserted by Theorem 1

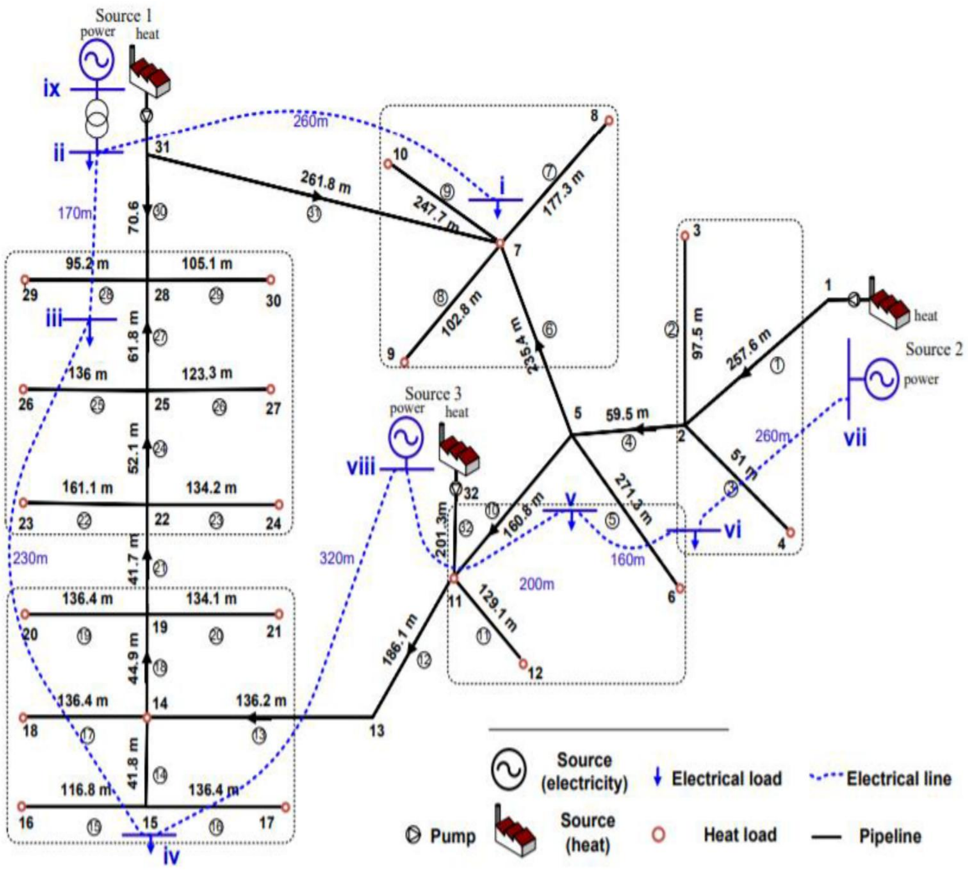

FIGURE 4 Schematic diagram of the electricity and district heating networks of the Barry Island case study

flexibility provided by the above three strategies, the criterion to screen flexibility providers is defined as follows:

$$C_{\text{flex}} = \sum_{\substack{i \neq j \\ i,j \in N_{WG}}} \frac{S_{ij}^{\text{improved}} - S_{ij}^{\text{original}}}{S_{ij}^{\text{original}}}$$

where S_{ij}^{original} represents the original area of the wind SOR before flexibility improvement on the projection of the wind injection i and j planes. S_{ij}^{improved} represents the area of the wind secure operation on the projection of the wind injection i and j planes when installing the additional flexibility provision. For N_{WG} wind generation, the flexibility is evaluated in the wind power injection region of every two wind generation. Then the flexibility of the total system can be quantified by aggregating the SOR of every two wind injection planes.

The evaluation criteria introduced in this article fully utilize the IES SOR to reflect the overall flexibility of the IES. The flexibility of the IES can be provided by introduced evaluation criterion in a global view. We notice that the existing flexibility metrics [20–22] only provide partial information of flexibility based on specific operational points or incomplete construction of SORs. Next, the additional flexibility providers; that is, installing the heat storage tank, installing the electric boiler, utilizing building thermal inertia, utilizing the plug-in hybrid electrical vehicle, or installing energy storage, are examined.

5.1 | Installing the electric boiler

The operational scheme for the electric boiler (EB) is designed to use wind power that may otherwise be curtailed. The additional electric load is added to the original power system, and simultaneously reduces the requirements of heat supply from the CHP plants. As a result, the CHP plants generate electricity more flexibly, and therefore better compensate the variability of wind power. The heat generated from the EB is linearly related to its consumption by a constant converting efficiency according to Equation (18):

$$\varphi_{\text{EB}}^{t} = \eta_{\text{EB}} P_{\text{EB}}^{t} \tag{18}$$

The operating cost of flexibility enhancement using the electric boiler is high. However, it has the largest potential for reducing the power output of the CHP plant. If the capacity of the electric boiler is large enough, the power output of a CHP plant can be reduced to zero, or even negative. Moreover, this strategy is robust to the variation of the wind power.

5.2 | Installing the heat storage tank

Another scheme to enhance the flexibility of CHP units for better accommodation of wind power is the usage of heat storage tanks. In these tanks, the interaction of flowing water with the district heating system changes the interior water temperature, providing a lower cost, more environmentally friendly energy storage system. Such devices could replace part of the heat production from CHP units, correspondingly reducing CHP power production when wind power is abundant. The formulation of heat storage tanks is expressed as follows:

$$\begin{aligned}&\varphi_{HS}^{t}=\varphi_{HS}^{t-1}+\Delta\varphi_{SC}^{t}-\Delta\varphi_{SD}^{t}\\&\varphi_{HS}^{\min}\le\varphi_{HS}^{t}\le\varphi_{HS}^{\max}\\&\Delta\varphi_{SC}^{\min}\le\Delta\varphi_{SC}^{t}\le\Delta\varphi_{SC}^{\max}\\&\Delta\varphi_{SD}^{\min}\le\Delta\varphi_{SD}^{t}\le\Delta\varphi_{SD}^{\max}\end{aligned}\tag{19}$$

The operating cost of flexibility enhancement by installing a heat storage tank is quite low. However, the effectiveness of flexibility enhancement using this strategy depends on the total heat stored in the heat storage tank.

5.3 | Utilizing building thermal inertia

Building thermal inertia is also an effective way to provide flexibility to system operations. The envelope structure of a building is able to store a certain amount of heat [21]. The indoor temperature decreases slowly when heating stops due to the thermal inertia of the maintenance structure, which allows buildings to be used as a heat storage unit with limited capacity. The thermal inertia of a building can be described in (20). To ensure a comfortable environment, the suitable range of indoor temperature that needs to be maintained is also expressed.

$$\begin{aligned}&T_{\text{in}}^{t}=T_{\text{in}}^{t-1}e^{-\Delta t/T_c}+\left(T_{\text{out}}^{t}+\varphi_i^t/\eta_{\text{air}}\right)\left(1-e^{-\Delta t/T_c}\right)\\&T_{\text{in}}^{\min}\le T_{\text{in}}^{t}\le T_{\text{in}}^{\max}\end{aligned}\tag{20}$$

Buildings are regarded as thermal storage units with a limited capacity to participate in CHP dispatching, which is more operable and economical than the extra installation of electric boiler and thermal storage tanks.

5.4 | Utilizing plug-in hybrid electrical vehicle

Flexibility in plug-in electric vehicle (PEV) charging can avoid renewable energy curtailment and improve the flexibility of the IES. The PEV charging is a natural source of load flexibility that can be properly controlled to provide demand response, thereby accommodating more renewable energy. The power requirement of PEVs can be shown as follows:

$$P_{\text{PEV},i,t}=\eta_{i,t}P_{\text{level},i}\tag{21}$$

The operating cost of flexibility improvement by utilizing plug-in hybrid electrical vehicle is quite low. The degree of flexibility enhancement depends on the total PEV charging.

5.5 | Installing energy storage

On the power grid side, the energy storage system has an advantage because of its rapid response feature, which can promote the consumption of large-scale renewable energy sources. The energy storage is a viable option to enhance the flexibility of the IES. The renewable energy can be stored during low-price periods and discharged back to the grid during high-price periods, higher profit can be achieved and the peak load can also be reduced to help alleviate transmission congestion. The formulation of energy storage is expressed as follows:

$$\begin{aligned}&E_i^{\min}\le\eta_iE_{i,t-1}+c_ip_{i,t}^{sc}\Delta T-\frac{1}{c_i}p_{i,t}^{sd}\Delta T\le E_i^{\max}\\&p_i^{sc\min}\le p_{i,t}^{sc}\le p_i^{sc\max}\\&p_i^{sd\min}\le p_{i,t}^{sd}\le p_i^{sd\max}\end{aligned}\tag{22}$$

The cost of installing energy storage is relatively high. The operation strategy of energy storage is quite important in achieving an optimal trade-off between the operation cost and revenue growth.

6 | NUMERICAL STUDY

Theorem 1 provides the theoretical foundation for complete characterization of the IES SOR. The QGS (8) is completely stable so that every trajectory converges to a stable equilibrium manifold. A computational scheme based on this complete stability property is developed to locate the path-connected feasible component. This scheme integrates the corresponding QGS, starting from any initial guess until it converges to its *w-limit point*, which is a feasible point. To illustrate the QGS-method, the Barry Island case is utilized to demonstrate it. Additionally, the IES SOR is utilized to analyse the impact of utilizing the plug-in hybrid electrical vehicle, installing energy storage, introducing the heat storage tank, electric boiler, and building thermal inertia on wind curtailment and flexibility enhancement of CHP units.

6.1 | The IES Barry Island system

The Barry Island electricity and district heating networks are shown in Figure 4. The heat network is a low temperature looped pipe district heating network fed by three CHP units.

The IES SOR of the Barry Island is characterized and shown in Figure 5. The proposed method is also applicable to completely characterize the IES SOR in three dimensions. Based on Theorem 1, visualizing the IES SOR can be achieved. The proposed QGS-based method can completely characterize the IES SOR through the regular stable equilibrium manifolds of the QGS system. When the operation point lies inside the IES SOR, it is a feasible operation point, even though it may touch the constraint boundary. However, when the operation point is outside the IES SOR, it violates operating constraints such as the voltage constraint, branch thermal limit and temperature drop equation.

6.2 | The flexibility enhancement of different strategies

Next, the flexibility improvement strategies on the power grid side and the heat network side will be discussed and compared respectively in the IES 9-5 test system. The IES 9-5 test system (Figure 6) has a 9-bus power system and a 5-node heating system. For the electric network, generators 1,3 are wind turbines and generator 2 is a CHP. For the heating network, generator 4 is modelled as a heating unit that has zero power capacity. The heat storage tank, the electric boiler, and the building thermal inertia are firstly combined with the CHP unit to evaluate the flexibility improvement. Utilizing the QGS-based method, the corresponding wind power injection region is constructed in Figure 7. Meanwhile, the results of the flexibility quantification of the test system using the CHP feasible operational region [22] are also shown in Figure 8. In this article, the feasible wind power injection region represents the (vector) amount of renewable energy that the power system can accommodate. This feasible wind power injection is deterministic and it reveals the allowable wind power that can be securely injected. The renewable uncertainties will be evaluated by the feasible wind power injection region.

We have the following observations:

1) The IES SOR is non-convex and contains multiple disconnected feasible components. It should be noted that the non-convex property of the IES SOR; hence, previous work [19] based on the convex assumption will give an overestimated result. The complete SOR of test system is characterized by proposed method, which lays the foundation for the further accurate flexibility quantification.
2) The IES SOR shrinks in size compared with the EPS SOR due to coupling with the heating network. Hence,

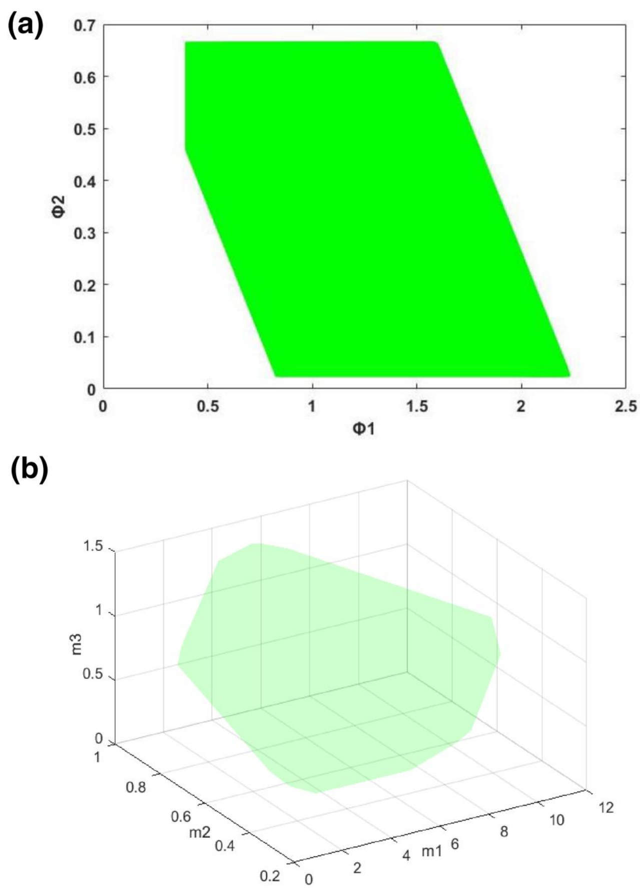

FIGURE 5 The IES SOR of the Barry Island is projected into (a) the Φ1×Φ2 space and (b) the $m_1 \times m_2 \times m_3$ space. Visualizing the IES SOR can be achieved through the regular stable equilibrium manifolds of the QGS system. When the operation point is within the IES SOR or on the boundary of the IES SOR, it indicates a feasible operation point. However, when the operation point is outside the IES SOR, it violates operating constraints such as the voltage constraint, the branch thermal limit, and the temperature drop equation

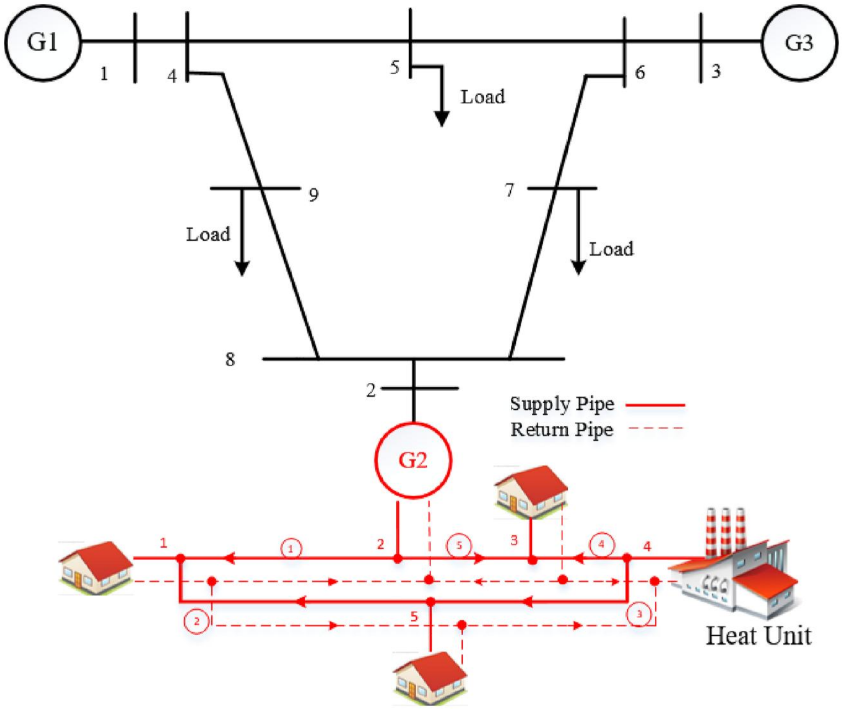

FIGURE 6 The IES 9-5 test system

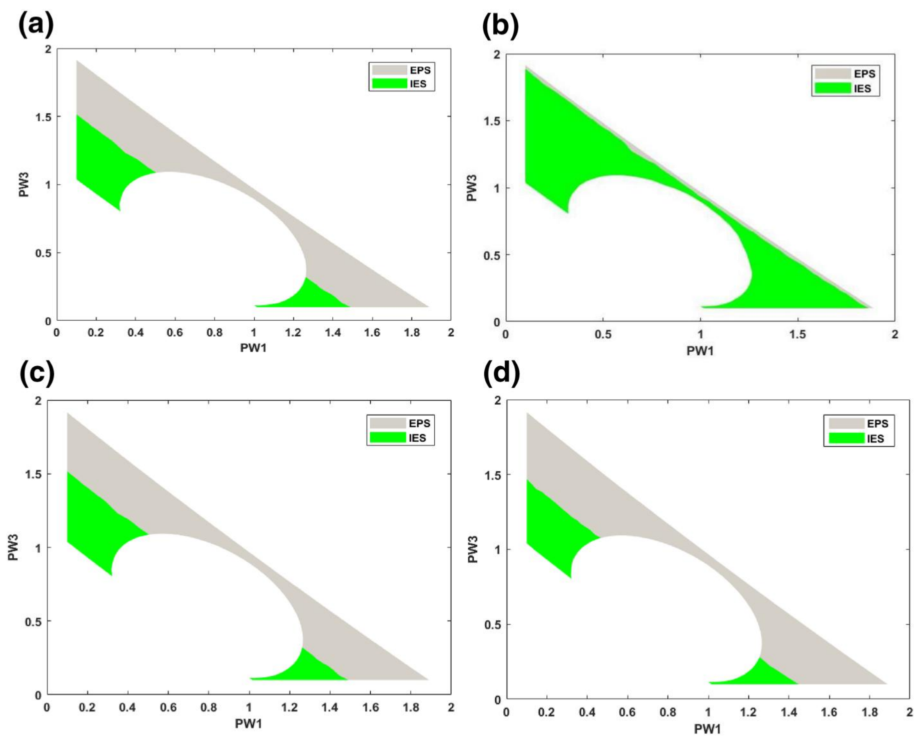

FIGURE 7 The feasible wind power injection region for the electric power system only (green) and with integrated heat-electricity (grey). (a) The feasible wind power injection region without additional heat sources. (b) The feasible wind power injection region with an installed electric boiler. (c) The feasible wind power injection region with an installed heat storage tank. (d) The feasible wind power injection region utilizing the building thermal inertia. The IES SOR shrinks in size compared with the EPS SOR due to coupling with the heating network. The marginal benefit of reducing the wind curtailment by the electric boiler is significant while the flexibility improvement by introducing heat storage and utilizing building thermal inertia is modest

combined heat and power (CHP) units with limited operational flexibility are a major barrier for accommodating more renewable energy.

3) For given operating conditions, the evaluation of flexibility improvement in the actual wind SOR instead of the previous CHP maximum feasible operational region

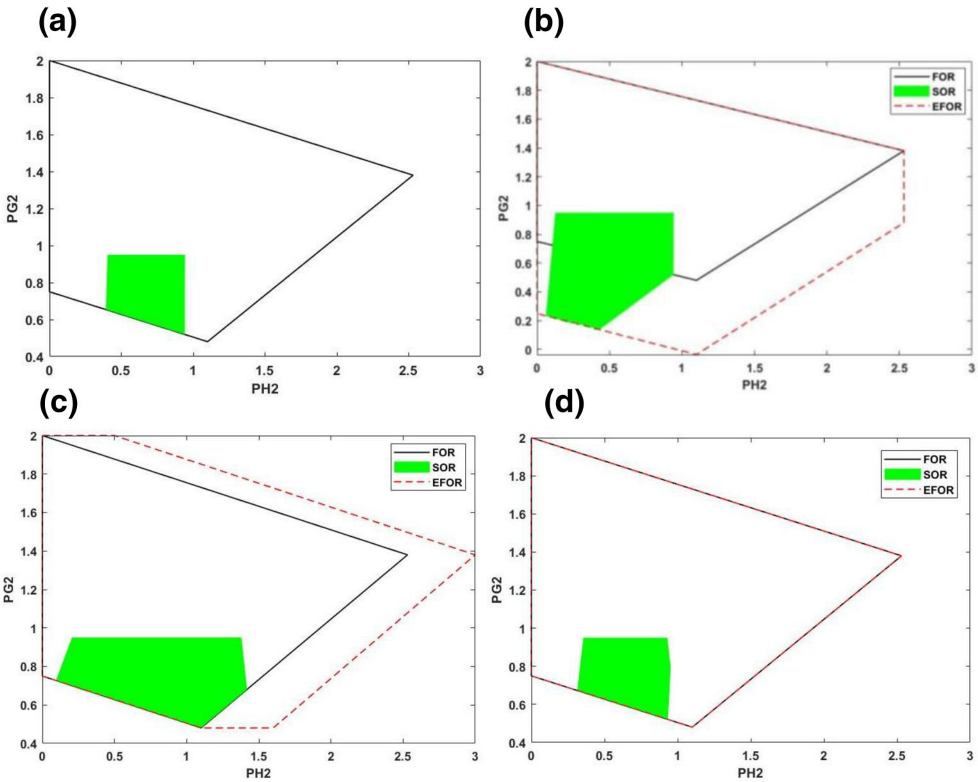

FIGURE 8 The black line represents the original boundaries of the CHP's maximum feasible operational region, the red dotted line represents the expanded boundaries of the CHP's maximum feasible operational region, whereas the green area shows the actual CHP SOR with respect to the current network. (a) without additional heat sources; (b) install an electric boiler; (c) install a heat storage tank; and (d) utilize building thermal inertia. Additional heat sources can expand the maximum production level of heat, whereas the minimum output level of electricity production decreases, which can promote the absorption of renewable energy in buses 1 and 3. Little improvement for feasibility enhancement is achieved by introducing heat storage and utilizing building thermal inertia, compared with installing an electric boiler. For the given operating conditions, the evaluation of flexibility improvement in the actual wind SOR instead of the previous maximum feasible operational region is more meaningful. However, from the perspective of the CHP's maximum feasible operational region, the heat storage tank will provide more flexibility than utilizing building thermal inertia; in fact, the flexibility improvement of the two, shown in actual wind secure operation, is almost the same

[22] is more meaningful. As can be observed in Fig. 8, the flexibility improvement by the electric boiler has obvious advantages over other strategies and the heat storage tank will provide more flexibility than utilizing building thermal inertia. Actually, when we observe the actual wind secure operation, installing the electric boiler indeed can provide more flexibility to IES while the flexibility improvement of installing the heat storage tank and utilizing building thermal inertia is almost the same. It reveals that the existing evaluation based on only the expansion of the CHP maximum feasible operational region provides partial insight. The evaluation criteria introduced in this article fully utilize the IES SOR to reflect the overall flexibility of the IES. Furthermore, the conducted numerical results show that these three strategies can provide improvement in the flexibility of the CHP and effectively reduces wind curtailment. In summary, the proposed SOR-based method is effective in quantifying the improvement in the flexibility of the CHP unit coupled to the heat storage tank, the electric boiler, and building thermal inertia

4) Installing the heat storage tank, the electric boiler, and utilizing building thermal inertia facilitates expanding the maximum production level of heat, whereas the minimum output level of electricity production decreases, which can promote the penetration of renewable energy in buses 1 and 3. Utilizing the proposed flexibility criteria, the flexibility enhancement of the IES coupling with heat storage tank, the electric boiler, and building thermal inertia can be quantified. The results show that little improvement for flexibility enhancement is achieved by introducing heat storage and utilizing building thermal inertia, which reduces the wind curtailment rate to 72.5% and 76.85% from 77.09%, respectively. However, the marginal benefit of reducing power production by the electric boiler is significant. When introducing the electrical boiler, the curtailment rate of wind power decreases to 8.69% of the total wind power production. Our introduced flexibility metrics screen out the flexibility

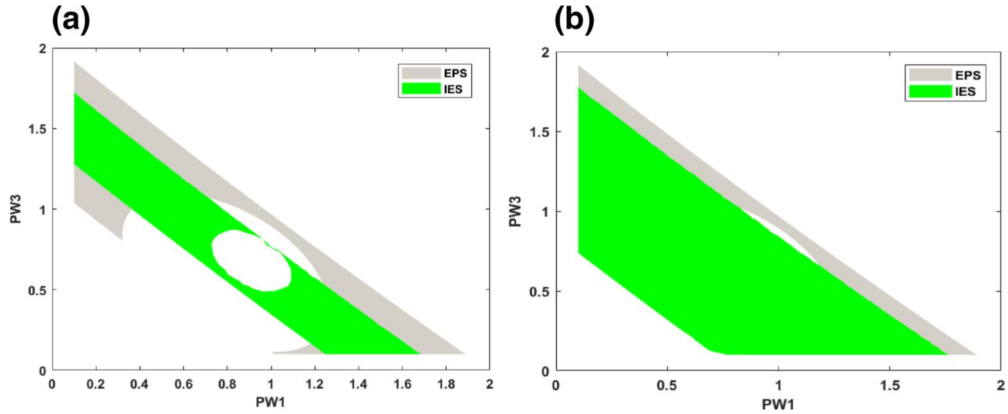

FIGURE 9 The feasible wind power injection region for the electric power system only (green) and with integrated heat-electricity (grey). (a) The feasible wind power injection region utilizing the plug-in hybrid electric vehicle. (b) The feasible wind power injection region installing energy storage. The marginal benefit of reducing the wind curtailment by energy storage is significant while the flexibility improvement by introducing the plug-in hybrid electric vehicle is modest

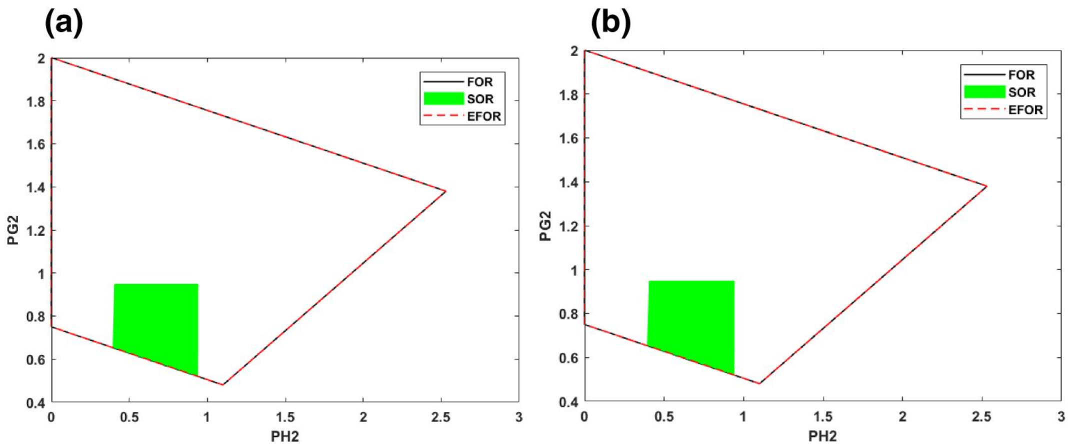

FIGURE 10 The black line represents the original boundaries of the CHP's maximum feasible operational region, the red dotted line represents the expanded boundaries of the CHP's maximum feasible operational region, whereas the green area shows the actual CHP SOR with respect to the current network. (a) utilizing the plug-in hybrid electric vehicle. (b) installing energy storage. From the perspective of the CHP's maximum feasible operational region, these two strategies have same effect on the flexibility improvement. In fact, the flexibility improvement of the two, shown in actual wind secure operation, is different

provision in a global view while previous quantification only based on specific operational points may provide partial flexibility improvement information of the whole system.

Next, the flexibility improvement strategies in the power grid side are evaluated in IES 9-5 test system. Utilizing the plug-in hybrid electric vehicle and installing energy storage are considered to promote renewable energy consumption. The wind power injection region is constructed in Figure 9. Meanwhile, the CHP feasible operational region in [22] is also characterized in Figure 10.

We have the following observations:

1) The IES SOR can be complete characterized by the proposed QGS-based method. Through the regular stable equilibrium manifold of the QGS, the SOR can be visualized. We notice that the property of non-convex can be observed.
2) From the perspective of CHP's maximum feasible operational region, utilizing the plug-in hybrid electric vehicle and installing energy storage have no effect on the CHP's feasible operational region. In fact, as can be seen in the feasible wind power injection region, the amount of renewable energy that can be integrated under different strategies is different. Installing energy storage can lead to more renewable energy absorption than utilizing the plug-in hybrid electric vehicle. Hence, it is more effective for flexibility quantification to observe the wind power injection region directly.
3) The complete SOR of the test system give us an overall view to analyse the IES flexibility. The performance of

installing energy storage is excellent, which can accommodate a wider range of wind energy, while the performance of utilizing the plug-in hybrid electric vehicle is not outstanding. The largest amount of the wind energy that the power system can consume under this strategy is greater than that of the IES without additional heat sources. However, the rate of the wind curtailment is still high. The complete IES SOR can provide flexibility enhancement information in a global view.

7 | CONCLUSION

There is a gap in the complete characterization for the SOR of the IES. To fill this gap, the exact AC power flow model and non-linear heating formulation without any approximation for the non-linear IES model are employed in our development for filling this gap. The complete characterization is derived in this article by developing the correspondence between the entire IES SOR and the union of regular stable equilibrium manifolds of the quotient gradient system derived from the exact IES model, and this correspondence is one-to-one. Our numerical results show that the IES SOR constructed by the proposed scheme provides exact characterization. This complete characterization advances the body of knowledge in describing the IES SOR. Moreover, the properties of non-convex and disjoint of the IES SOR can be observed by the proposed approach. The derived complete characterization significantly improves the existing methods which often produce overestimated or conservative estimation.

To reconcile the conflict between inflexible CHP operations and the full wind power injection, there are different strategies to achieve additional flexibility, such as utilizing the plug-in hybrid electric vehicle and installing energy storage, installing the heat storage tank, the electric boiler, and building thermal inertia. The operational flexibility improvement by these strategies is examined from the SOR perspective. The exact IES SOR lays the foundation for the accurate flexibility quantification of the IES. The flexibility quantification metric derived from the IES SOR is presented to evaluate the effectiveness of different strategies in enhancing the IES flexibility. Numerical studies show that the proposed evaluation criterion based on the complete IES SOR is effective to identify the most effective flexibility providers. However, the previous evaluation methods based on only specific operational points or incomplete SOR may provide partial or even inaccurate information regarding the IES flexibility. Of the five strategies, the marginal benefit of reducing the wind curtailment by the electric boiler is significant. Second, installing energy storage is also effective in flexibility enhancement. Our proposed metric is useful to identify the most effective flexibility providers. Our future work will focus on the fast computation of the IES SOR for real-time application.

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Appendix

Proof of Theorem 1

Proof: To prove this theorem, we will show that the quotient gradient system (17) admits the function $E(x) = 1/2\|H(x)\|^2$ which is the energy function of the quotient gradient system (17). The function *E(x)* satisfies the following three conditions of energy functions (E1), (E2), and (E3) [23,24].

(E1) $\dot{x} = 0$ implies that $\nabla E = 0$.
(E2) The function along any non-trivial trajectory $\Phi(t, x_0)$ is non-increasing, that is, $dE(\Phi(t, x_0))/dt \le 0$ and the set $\{t \in \Re : E(\Phi(t, x_0)) = 0\}$ has measure zero.
(E3) If along any trajectory $\{E(\Phi(t, x_0)) : t \ge 0\}$ is bounded, then the trajectory $\{\Phi(t, x_0) : t \ge 0\}$
First, voltage angle constraints are added to constraint set (16):

$$\theta_i^2 \le \left(\theta_i^{\max}\right)^2 \quad i = \{1, \ldots, N_B\}$$

where $\theta_i^{\max}$ is usually chosen as *2π*, which does not affect the final solutions of constraint set (16) and will help avoid periodic solutions.

By adding slack variables to each inequality constraint, the constraint functions can be defined:

$$H(x) = \begin{bmatrix} P_G - P_L - P(\theta,\ V) \\ Q_G - Q_L - Q(\theta,\ V) \\ \theta^2 - (\theta^{\max})^2 + SV_\theta^2 \\ S_f^2 - \left(S_l^{\max}\right)^2 + SV_{Sf}^2 \\ S_t^2 - \left(S_l^{\max}\right)^2 + SV_{St}^2 \\ V - V^{\max} + SV_{Vh}^2 \\ V^{\min} - V + SV_{Vl}^2 \\ P_G - P_G^{\max} + SV_{PGh}^2 \\ P_G^{\min} - P_G + SV_{PGl}^2 \\ Q_G - Q_G^{\max} + SV_{QGh}^2 \\ Q_G^{\min} - Q_G + SV_{QGl}^2 \\ Am_l - m_{qi} \\ BKm_{qi}|m_{qi}| \\ \phi_i - C_p m_{qi}(T_{si} - T_{oi}) \\ T_{ei} - (T_{si} - T_a)e^{-\frac{\lambda L}{C_p m_i}} - T_a \\ \alpha_1 P_H + \beta_1 \phi_H + SV_{chph}{}^2 \\ \alpha_2 P_H + \beta_2 \phi_H + SV_{chpl1}{}^2 \\ \alpha_3 P_H + \beta_3 \phi_H + SV_{chpl2}{}^2 \end{bmatrix} = \begin{pmatrix} H1' \\ H2' \\ H3' \\ H4' \\ H5' \\ H6' \\ H7' \\ H8' \\ H9' \\ H10' \\ H11' \\ H12' \\ H13' \\ H14' \\ H15' \\ H16' \\ H17' \\ H18' \end{pmatrix} = 0$$

$$H(x) = \begin{bmatrix} m_i - m^{\max} + SV_{mh}{}^2 \\ m^{\min} - m_i + SV_{ml}^2 \\ T_{si} - T_s{}^{\max} + SV_{Tsh}^2 \\ T_s{}^{\min} - T_{si} + SV_{Tsl}^2 \\ T_{ri} - T_r{}^{\max} + SV_{Trh}^2 \\ T_r{}^{\min} - T_{ri} + SV_{Trl}^2 \\ \phi_s - \phi^{\max} + SV_{hh}^2 \\ \phi^{\min} - \phi_s + SV_{hl}^2 \end{bmatrix} = \begin{pmatrix} H19' \\ H20' \\ H21' \\ H22' \\ H23' \\ H24' \\ H25' \\ H26' \end{pmatrix} = 0$$

By differentiating *E(x)*, one obtains:

$$\frac{\partial E(x)}{dt} = \langle \nabla E(x), \dot{x} \rangle = \langle DH(x)^T \cdot H(x), -DH(x)^T \cdot H(x) \rangle$$
$$= - \left\| DH(x)^T \cdot H(x) \right\|^2 \leq 0$$

Therefore, condition (E1) holds. Clearly, $\frac{\partial E(x)}{dt}$ if and only if x is an equilibrium point. Thus, condition (E2) is also satisfied. It remains to be shown that condition (E3) is valid. To this end, we will prove the inverse negative proposition of (E3), that is, If $\|x\| \to \infty$ then $E(x) = \frac{1}{2}\|H(x)\|^2 \to \infty$.

Observe that if $\|x\| \to \infty$, then at least one variable in set $\{\theta, V, P_G, Q_G, T_s, T_r, P_H, \phi_H, m, SV\}$ goes to infinity. In this regard, we discuss the following cases.

Case 1: If one or more variables in $\{\theta, SV_\theta, SV_{Sf}, SV_{St}\}$ go to infinity, then one can easily check that either (H3'), (H4'), or (H5') must become unbounded because $\theta^{\max}$ and $S_l^{\max}$ are fixed constants, while the terms $\theta^2, SV_\theta^2, SV_{Sf}^2$, and SV_{St}^2 are all non-negative. This shows that $E(x)$ becomes unbounded if one or more variables in $\{\theta, SV_\theta, SV_{Sf}, SV_{St}\}$ go to infinity.

Case 2: If one or more variables in the complement $\{V, SV_{Vh}, SV_{Vl}, P_G, SV_{PGh}, SV_{PGl}, Q_G, SV_{QGh}, SV_{QGl}, m, SV_{mh}, SV_{ml}, T_s, SV_{Tsh}, SV_{Tsl}, T_r, SV_{Trh}, SV_{Trl}, \varphi_H, SV_{hh}, SV_{hl}, P_H, SV_{chph}, SV_{chpl}\}$ go to infinity, the same conclusion can be derived as follows. To fix the ideas, one can start by considering the case in which one or more variables in $\{V, SV_{Vh}, SV_{Vl}\}$ go to infinity. Indeed, if one or more variables in $\{V, SV_{Vh}, SV_{Vl}\}$ go to infinity, the discussion can be further sub-divided below for the sake of clarity.

Case 2.1: If V goes to positive infinity, one observes that (H6') must become unbounded (positive infinity), since $V^{\max}$ is constant and SV_{Vh}^2 is non-negative. Similarly, (H7') becomes unbounded (positive infinity) if V goes to negative infinity. In other words, $E(x)$ always becomes unbounded, regardless of whether V goes to positive or negative infinity.

Case 2.2: If V is bounded but one or more variables in $\{SV_{Vh}, SV_{Vl}\}$ go to infinity, then (H6') becomes unbounded as SV_{Vh} goes to infinity, since V is bounded and $V^{\max}$ is constant. Similarly, (H7') becomes unbounded as SV_{Vl} goes to infinity. Therefore, *E(x)* becomes unbounded if either SV_{Vh} or SV_{Vl} goes to infinity.

To summarize, the value of *E(x)* becomes unbounded if one or more variables in $\{V, SV_{Vh}, SV_{Vl}\}$ go to infinity. By analogy, the cases for one or more variables in $\{P_G, SV_{PGh}, SV_{PGl}\}, \{Q_G, SV_{QGh}, SV_{QGl}\}, \{m, SV_{mh}, SV_{ml}\}, \{T_s, SV_{Tsh} SV_{Tsl}\}, \{T_r, SV_{Trh}, SV_{Trl}\}, \{\varphi_H, SV_{hh}, SV_{hl}\}$, *or* $\{P_H, SV_{chph}, SV_{chpl}\}$ that go to infinity can be similarly analysed. The above discussion leads to the conclusion that *E(x)* becomes unbounded as well if $\|x\| \to \infty$. Hence, condition (E3) is verified. Above all, *E(x)* satisfies conditions (E1)–(E3) and is an energy function for system (17).

First, we prove the necessity. Assume SSR^i is a feasible component of constraint set (17) and $x^*$is a feasible point on SOR^i, that is, $x^* \in SOR^i, H^n(x^*) = 0$.

Then $-DH^n(x^*)^T H^n(x^*) = 0$ is also satisfied. Thus, x^* also belongs to the equilibrium manifold of system (17). Now the main task is to verify whether x^* belongs to a stable equilibrium manifold of (17), since the energy function

$$E(x) = \tfrac{1}{2}\|H^n(x(t))\|^2 \geq 0 \quad \text{and } E(x^*) = \tfrac{1}{2}\|H^n(x^*)\|^2 = 0.$$

This implies that $E(x^*)$ achieves the global minimum of manifolds Σ_i^{rs} for some $i \in J = \{1, ..., n\}$, i.e., $SOR^i = \Sigma_i^{rs}$. In other words, the entire feasible region of constraint set (16) is contained in the union of the stable equilibrium manifolds of QGS (17), that is, $SSR^i \in \cup \Sigma_i^{rs}$ and $H(SOR^i) = 0$ holds. Thus, according to Definition 3, SOR^i is a regular stable equilibrium manifold.

Next, we show the sufficiency. Assume Σ_i^{rs} is a regular stable equilibrium manifold of QGS (17). Then, according to its definition, $H(\Sigma_i^{rs}) = 0$, which means that Σ_i^{rs} is a feasible component of constraint set (16). This completes the proof.